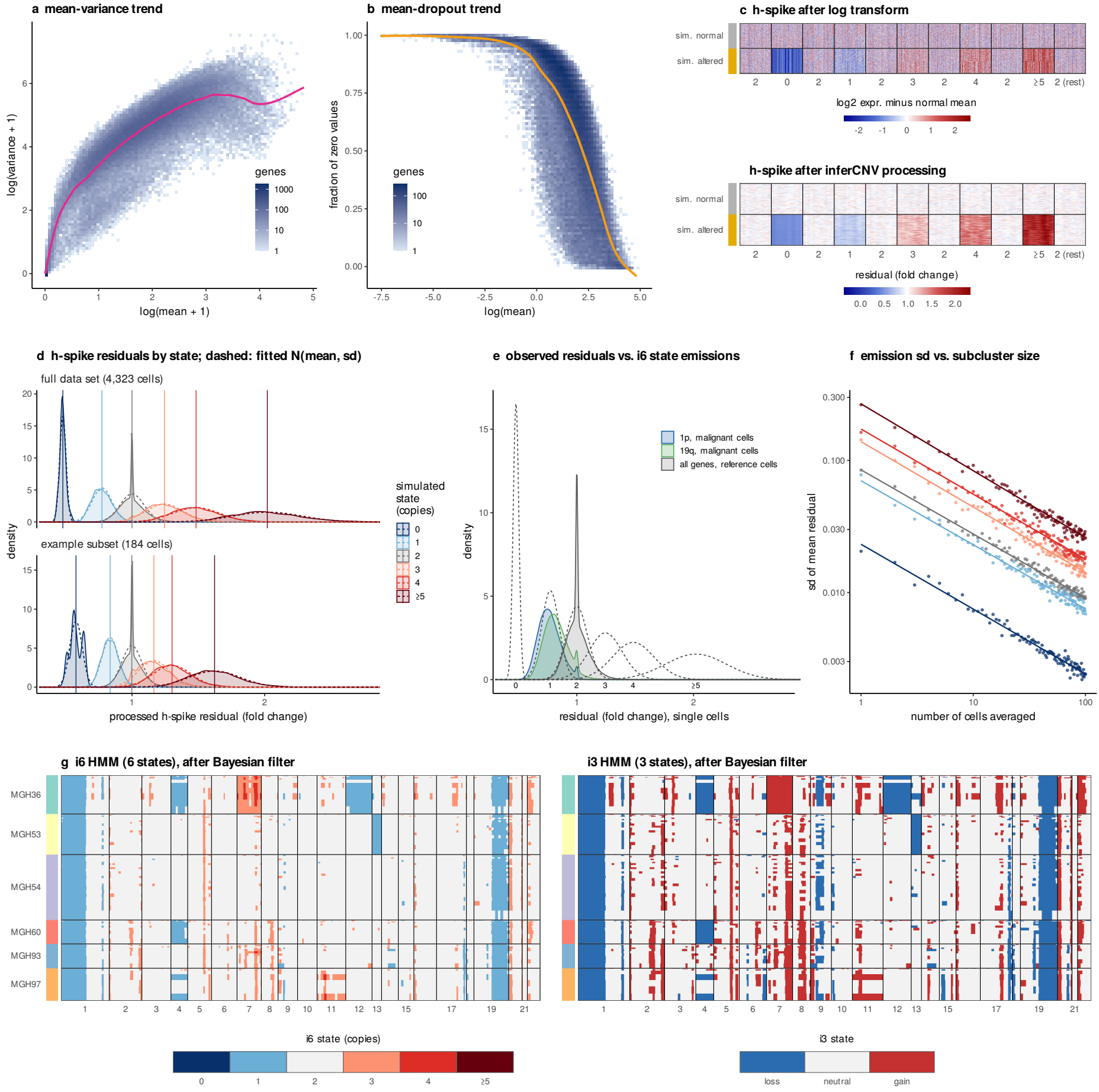

a mean-variance trend
log(variance + 1)
log(mean + 1)
genes
1000
100
10
1
b mean-dropout trend
fraction of zero values
log(mean)
genes
100
10
1
c h-spike after log transform
sim. normal
sim. altered
2 0 2 1 2 3 2 4 2 ≥5 2 (rest)
log2 expr. minus normal mean
-2 -1 0 1 2
h-spike after inferCNV processing
sim. normal
sim. altered
2 0 2 1 2 3 2 4 2 ≥5 2 (rest)
residual (fold change)
0.0 0.5 1.0 1.5 2.0
d h-spike residuals by state; dashed: fitted N(mean, sd)
full data set (4,323 cells)
example subset (184 cells)
density
processed h-spike residual (fold change)
simulated state (copies)
0
1
2
3
4
≥5
e observed residuals vs. i6 state emissions
density
residual (fold change), single cells
1p, malignant cells
19q, malignant cells
all genes, reference cells
f emission sd vs. subcluster size
sd of mean residual
number of cells averaged
g i6 HMM (6 states), after Bayesian filter
MGH36
MGH53
MGH54
MGH60
MGH93
MGH97
i6 state (copies)
0 1 2 3 4 ≥5
i3 HMM (3 states), after Bayesian filter
i3 state
loss
neutral
gain